The extremal types theorem and the GEV

2026-09-14

Summary

The GEV distribution, where it comes from, and how to estimate its parameters.

- Block maxima of any iid sequence, suitably normalized, converge to a GEV.
- The GEV has three parameters: location μ, scale $\sigma > 0$, and shape ξ.
- $\xi > 0$: heavy tail, unbounded (Fréchet). $\xi = 0$: exponential tail (Gumbel). $\xi < 0$: bounded upper tail (Weibull).
- The T-year return level for the Gumbel is $z_T = \mu - \sigma \ln[-\ln(1 - 1/T)]$.
- MLE finds the parameters that make the data most probable; L-moments are an alternative that is less sensitive to outliers.
- Bayesian inference puts priors on (μ, σ, ξ) and samples the posterior, so uncertainty propagates automatically to return levels.

Key results.

$$F(z) = \exp\left\{ -\left[1 + \xi\left(\frac{z - \mu}{\sigma}\right)\right]^{-1/\xi} \right\}$$

$$z_T = \mu - \frac{\sigma}{\xi}\left\{ 1 - \left[-\ln\left(1 - \frac{1}{T}\right)\right]^{-\xi} \right\}$$

$$p(\mu, \sigma, \xi \mid y) \propto \prod_{i=1}^{m} f_{\mathrm{GEV}}(y_i \mid \mu, \sigma, \xi)\, p(\mu, \sigma, \xi)$$

Bibliography